INCH-POUND

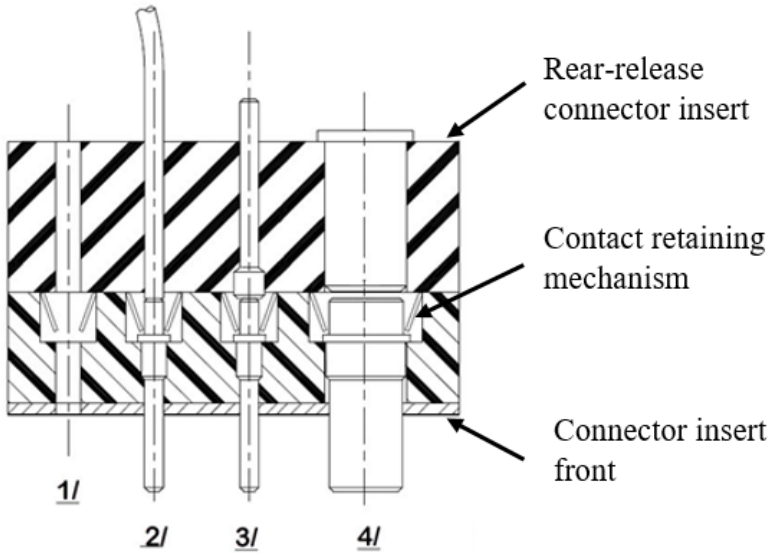
MS27488P
22 October 2020
SUPERSEDING
MS27488N
22 July 2008

DETAIL SPECIFICATION SHEET

PLUG, END SEAL, ELECTRICAL CONNECTOR

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.



NOTES:
1/ Typical connector cavity without sealing plug.
2/ Typical wired contact installed in connector cavity.
3/ Unwired contact and sealing plug MS27488-XX-2, or -3.
4/ Unwired contact and optional design sealing plug MS27488-0-3.

FIGURE 1. Sealing plug configurations.

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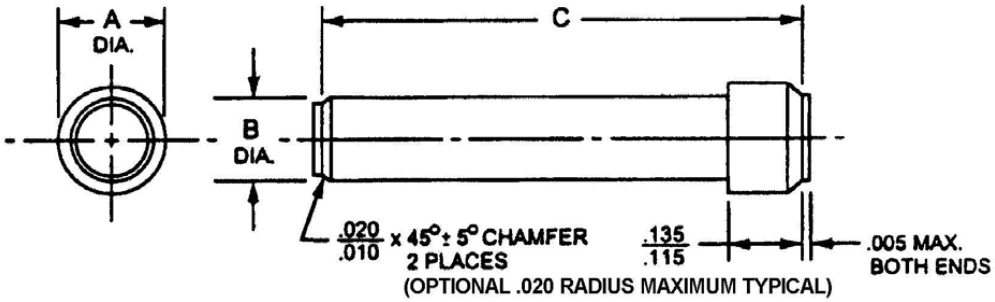


FIGURE 2. MS27488-XX-2 or -3 sealing plug.

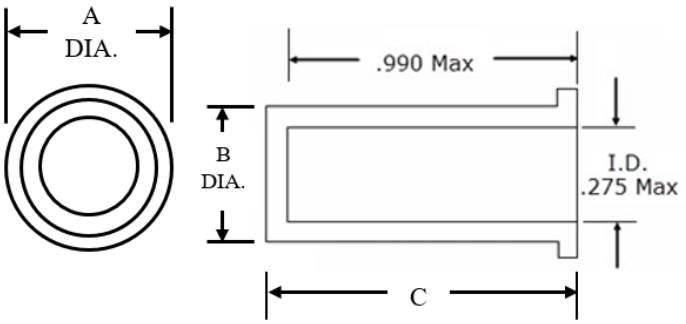


FIGURE 3. Optional design for MS27488-0-3.

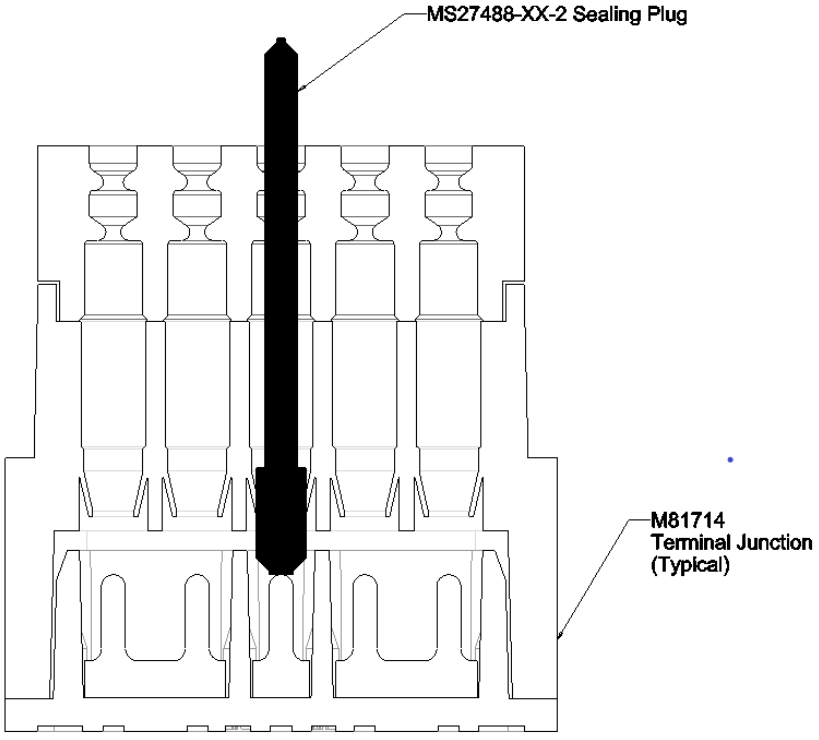


FIGURE 4. Sealing plug installation in a terminal junction module (SAE AS81714/XX).

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TABLE I. MS27488-2 and -3 Sealing plug (dimensions and application information).

MS27488		Color code	A dia. 2/	B dia. 2/	C dim. 2/	Applications 3/	Supersession
Gauge size	Dash no. 1/						
-22 4/	-2	Black	.066 (1.676) .058 (1.473)	.045 (1.143) .038 (0.965)	.668 (16.970) .618 (15.697)	MIL-DTL-26500, MIL-DTL-28804, SAE AS29600, MIL-DTL-38999, SAE-AS81511 Series 3 & 4, and SAE-AS81659, SAE-AS81714	MS27488-22-1, M81511/39-22 and MS27488-22, MS27488-A22
-20	-2	Red	.090 (2.286) .080 (2.032)	.058 (1.473) .048 (1.219)	.805 (20.447) .755 (19.177)	MIL-DTL-26482 Series 1 & 2, MIL-DTL-26500, MIL-DTL-26518, SAE AS29600, MIL-DTL-38999 ALL Series, MIL-PRF-22885/108, MIL-S-24317/12, MS21346, MS21347, MS27784, MS27785, MS27786, MS27787, MS27788, MS27789, MIL-DTL-83527, MIL-DTL-83723 Series II & III, SAE-AS81659, SAE-AS81703, SAE-AS81714, SAE-AS81582/2, and SAE-AS81511 Series 3 & 4	MS27488-20-1, M81511/39-20, M83723/28-20, MS27186-1, MS27187-3, and MS27488-20 MS3187-20-2, MS27488-A20
-16	-2	Green	.130 (3.302) .120 (3.048)	.079 (2.007) .069 (1.753)	.872 (22.149) .822 (20.879)	MIL-DTL-26482 Series 1 & 2, MIL-DTL-26500, MIL-DTL-26518, SAE AS29600, MIL-DTL-38999 ALL Series, MS27722, MS27723, MS27724, MS27781, MS27782, MS27783, MIL-DTL-83527, MIL-DTL-83538, MIL-DTL-83723 Series II & III, SAE-AS81703, SAE-AS81659, SAE-AS81714, SAE-AS81582/2, and SAE-AS81511 Series 3 & 4,	M81511/39-16, M83723/28-16, MS27186-2, MS27187-1, MS27488-16-1, and MS27488-16, MS3187-16-2, MS27488-A16
-12	-2	Orange	.170 (4.318) .160 (4.064)	.125 (3.175) .115 (2.921)	.841 (21.361) 0.791 (20.091)	MIL-DTL-26482 Series 1 & 2, MIL-DTL-26500, MIL-DTL-26518, SAE AS29600, MIL-DTL-38999 ALL Series, MIL-DTL-83527, MIL-DTL-83538, MIL-DTL-83723 Series II & III, SAE-AS81582/2, SAE-AS81714, SAE-AS81703, SAE-AS81659, and SAE-AS81511 Series 3 & 4,	M81511/39-12, M83723/28-12, MS27186-3, MS27187-2, MS27488-12-1 and MS27488-12, MS3187-12-2, MS27488-A12
-16	-3	Blue	.130 (3.302) .120 (3.048)	.079 (2.007) .069 (1.753)	1.472 (37.389) 1.422 (36.119)	SAE-AS50151	MS27488-16-1 and MS27488-16
-12	-3	Yellow	.170 (4.318) .160 (4.064)	.125 (3.175) .115 (2.921)	1.441 (36.601) 1.391 (35.331)	SAE-AS50151	MS27488-12-1 and MS27488-12
-8	-3	Red	.315 (8.001) .305 (7.747)	.175 (4.445) .165 (4.191)	1.189 (30.201) 1.139 (28.931)	SAE-AS50151, MIL-DTL-26500	MS27488-8-1, M83723/28-8, and MS27488-8, MS3187-8-2
-4	-3	Blue	.414 (10.516) .404 (10.262)	.283 (7.188) .273 (6.934)	1.189 (30.201) 1.139 (28.931)	SAE-AS50151	MS27488-4-1, M83723/28-4, MS27187-4, and MS27488-4, MS3187-4-2
-0 5/	-3	Yellow	.606 (15.392) .596 (15.138)	.431 (10.947) .431 (10.947)	1.071 (27.203) 1.021 (25.933)	SAE-AS50151	MS27488-0-1, M83723/28-0, and MS27488-0, MS3187-0-2

NOTES:
1/ -2 is a short sealing plug; -3 is a long sealing plug (see figure 2).
2/ Dimensions are in inches (millimeters); Metric equivalents to the nearest 0.001 are given for general information and for reference only; Values are calculated and rounded on 1 inch = 25.4 millimeters.
3/ See Table II.
4/ For connector cavity sizes 23-22.
5/ See figure 3.

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TABLE II. Components that use MS27488 sealing plugs.

Component	Component type	Sealing Plug Dash No.
SAE-AS50151	Electrical connector	MS27488-XX-3
MIL-PRF-22885/108	Pushbutton switch	MS27488-XX-2
MIL-S-24317/12	Pushbutton switch	MS27488-XX-2
MIL-DTL-26482, Series 1 and 2	Electrical connector	MS27488-XX-2
MIL-DTL-26500	Electrical connector	MS27488-XX-3
MIL-DTL-26518	Electrical connector	MS27488-XX-2
MIL-DTL-28804	Electrical connector	MS27488-XX-2
SAE-AS29600	Electrical connector	MS27488-XX-2
MIL-DTL-38999	Electrical connector	MS27488-XX-2
SAE-AS81511, Series 3 and 4	Electrical connector	MS27488-XX-2
SAE-AS81582/2	Electrical connector	MS27488-XX-2
SAE-AS81659	Electrical connector	MS27488-XX-2
SAE-AS81703	Electrical connector	MS27488-XX-2
SAE-AS81714	Terminal Junction System	MS27488-XX-2
SAE-AS81714/1 - /9	Terminal Junction System	MS27488-XX-2
SAE-AS81714/17	Terminal Junction System	MS27488-XX-2
SAE-AS81714/25 - /28	Terminal Junction System	MS27488-XX-2
SAE-AS81714/30 - /31	Terminal Junction System	MS27488-XX-2
SAE-AS81714/60 - /61	Terminal Junction System	MS27488-XX-2
SAE-AS81714/63	Terminal Junction System	MS27488-XX-2
MIL-DTL-83527	Electrical connector	MS27488-XX-2
MIL-DTL-83723, Series II and III	Electrical connector	MS27488-XX-2
MIL-DTL-83723/66 - /69	Electrical connector	MS27488-XX-2
MS21346	Toggle switch	MS27488-XX-2
MS21347	Toggle switch	MS27488-XX-2
MS27722	Toggle switch	MS27488-XX-2
MS27723	Toggle switch	MS27488-XX-2
MS27724	Toggle switch	MS27488-XX-2
MS27781	Toggle switch	MS27488-XX-2
MS27782	Toggle switch	MS27488-XX-2
MS27783	Toggle switch	MS27488-XX-2
MS27784	Toggle switch	MS27488-XX-2
MS27785	Toggle switch	MS27488-XX-2
MS27786	Toggle switch	MS27488-XX-2
MS27787	Toggle switch	MS27488-XX-2
MS27788	Toggle switch	MS27488-XX-2
MS27789	Toggle switch	MS27488-XX-2

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REQUIREMENTS:

1. Material. Plastic rod, molded and extruded - temperature limit 200 °C minimum. The material used to fabricate sealing plugs shall be polytetrafluoroethylene (PTFE) in accordance with ASTM-D1710, or fluorinated ethylene propylene (FEP) in accordance with ASTM-D2116, or perfluoroalkoxy (PFA) in accordance with ASTM-D3307, or polyphenylsulfone (PPSU) in accordance with ASTM-D6394.

2. Color. Color of items shall be in accordance with SAE-AMS-STD-595.

3. Resistivity. Material resistivity shall be not less than 100 megohms when tested in accordance with ASTM-D257.

4. Workmanship. The item shall be free from flash and burrs. The length of diameter C shall be free of all molding marks, protrusions, and blemishes. Molding marks, protrusions, and blemishes from gate cutoffs are permissible but only on the plug ends or side of the head and shall not be greater than .005 inch in height.

5. Part marking. Marking is not required. Items shall be identified at the bag or higher packaging level.

6. Connector, module, and switch manufacturers, their distributors, and agents supplying military connectors, modules, and switches as noted, shall only furnish seal plugs with the connectors, modules, and switches in accordance with this specification sheet.

7. Installation.

a. When installing sealing plugs in connectors, components, or terminal junction module grommets, the head of the sealing plug shall be inserted first.

b. When crimp contact connectors are used, the unused contacts shall be installed in unused cavities. MS27488 or applicable sealing plugs shall be inserted in unused grommet holes of environment resistant connectors. Sealing plugs shall not be installed where they interfere with the free movement of the spring-loaded socket termini on connectors containing fiber optic termini.

c. For installing sealing plugs into grommets of connectors, or components, the sealing plug is installed after an unwired contact is inserted in the unused connector cavity applications. The sealing plug shall be pushed into the grommet until the head is positioned against the end of the unwired contact (see figure 1).

d. For installing sealing plugs into grommets of SAE AS81714/XX terminal junction modules, or inline splices, the sealing plug is installed without an unwired contact installed. The sealing plug shall be pushed into the grommet until the head is positioned against the contact retention system and bottoms out (see figure 4).

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NOTES:

1. Application: The sealing plugs defined in this specification are intended to be inserted in applicable electrical connectors and terminal junction modules which have unfilled or unused cavities (see figures 1 and 4).

2. There are several MS27488 sealing plug configurations, they are defined by contact size and application requirements (see tables I and II). Once installed in the intended connector, the sealing plug is intended to seal the open wire cavity and prevent fluid or contaminant entry into the connector. Failure to employ a sealing plug, or the correct size or configuration sealing plug, may result in connector shorting out, system degradation or failure, or injury to personnel.

3. Dimensioning and tolerancing per ASME Y14.5M.

4. Table II lists the documents that use MS27488 sealing plugs and table III identifies their status and superseding document information.

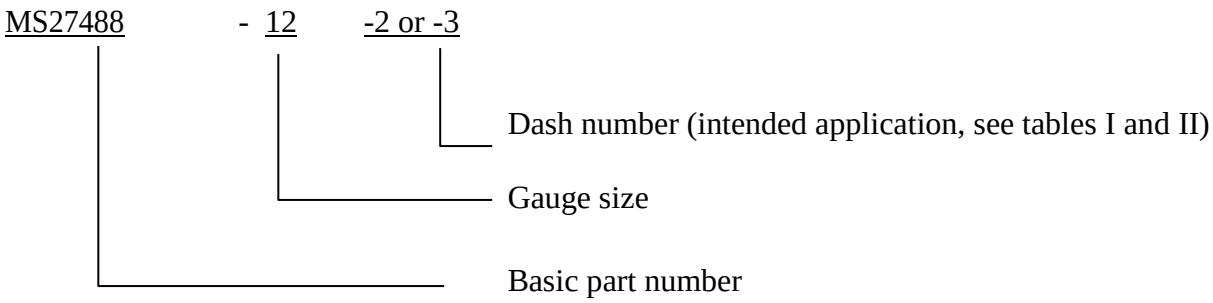
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TABLE III. Document status and supersession information.

Document	Status	Superseding Document
MIL-DTL-5015	Canceled	SAE-AS50151
MIL-W-5088	Inactive for New Design	Use SAE-AS50881
MIL-PRF-22885/108	Active	None
MIL-S-24317/12	Inactive for New Design	None
MIL-DTL-26482, Series 1 and 2	Active	None
MIL-DTL-26500	Inactive for New Design	MIL-DTL-83723, series III
MIL-DTL-26518	Inactive for New Design	None
MIL-DTL-28804	Active	None
MIL-C-29600	Canceled	SAE-AS29600
MIL-DTL-38999	Active	None
MIL-C-81511, Series 3 and 4	Canceled	SAE-AS81511
MIL-C-81582/2	Canceled	SAE-AS81582/2
MIL-C-81659	Canceled	SAE-AS81659
MIL-T-81714	Canceled	SAE-AS81714
MIL-T-81714/1 - /9	Canceled	SAE-AS81714/1 - /9
MIL-T-81714/17	Canceled	SAE-AS81714/17
MIL-T-81714/25 - 28	Canceled	SAE-AS81714/25 - 28
MIL-T-81714/30 - /31	Canceled	SAE-AS81714/30 - /31
MIL-T-81714/60 - /61	Canceled	SAE-AS81714/60 - /61
MIL-T-81714/63	Canceled	SAE-AS81714/63
MIL-DTL-83527	Active	None
MIL-DTL-83538	Canceled	None
MIL-DTL-83723, Series II and III	Active	None
MIL-DTL-83723/66 - /69	Active	None
MS21346	Active	None
MS21347	Active	None
MS27722	Active	None
MS27723	Active	None
MS27724	Active	None
MS27781	Active	None
MS27782	Active	None
MS27783	Active	None
MS27784	Active	None
MS27785	Active	None
MS27786	Active	None
MS27787	Active	None
MS27788	Active	None
MS27789	Active	None
MIL-STD-1760	Active	None

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5. Example of part or identifying number:



6. The MS27488 gauge size given as part of the part number matches the size of the contact crimp barrel.

7. For design purposes, this specification sheet takes precedence over acquisition documents referenced herein.

8. Tooling. Contact installing tools listed in table IV may be used for installing sealing plugs. The use of tools for sealing plug sizes 12 and smaller will aid in the installation by supporting the plug during headfirst insertion. The tool rides against the shoulder on the back of the plug head during insertion rather than pushing on the tail end of the unsupported plug. The tools will not remove the sealing plugs. Sealing plugs must be removed by grasping the tail end of the sealing plug that is left protruding outside the grommet. The tools are not recommended for sealing plug sizes 8 and larger.

TABLE IV. Installation tools.

Seal plug	Tool part number
MS27488-22-2	M81969/14-01
MS27488-20-2	M81969/14-11
MS27488-16-2	M81969/14-03
MS27488-12-2	M81969/14-04
MS27488-16-3	M81969/14-03
MS27488-12-3	M81969/14-04

9. Manufacturers are permitted to supply MS27488, revision N sealing plugs for 12 months after the date of the MS27488, revision P or until they have exhausted their inventories, whichever comes first.

10. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

(Project 5935-2020-103)

Review activities:
Army - AR, MI
Navy - EC
Air Force - 19

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